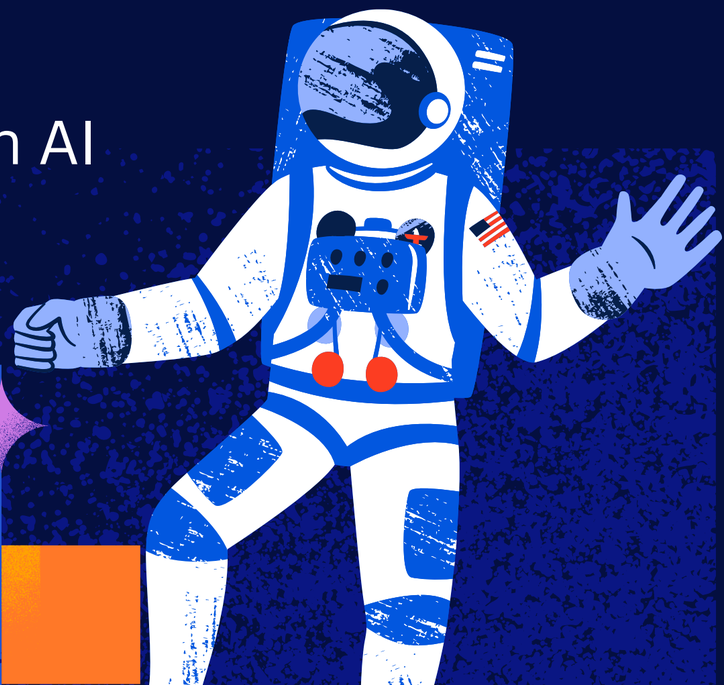
NASA INTERNATIONAL SPACE APPS CHALLENGE

ExoVision by Stellar Detectives

A World Away: Hunting for Exoplanets with AI

By Abdullah, Saad, Muhammad, Taimoor





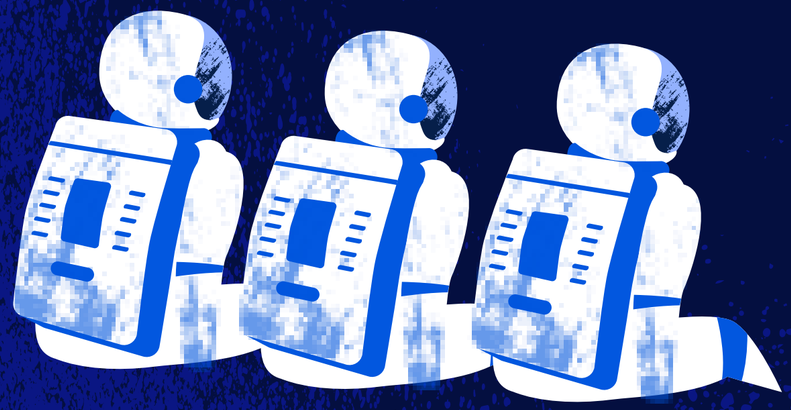
A Universe of Data. A Needle in the Stars.

- NASA’s Kepler, K2, and TESS missions collect billions of starlight readings.
- Hidden within them are faint dips — signs of unseen planets.
- Manually detecting these signals is like finding a needle in a galaxy-sized haystack.

Transforming Light into Discovery

ExoVision is an AI-driven platform that analyzes NASA’s mission datasets — automatically detecting exoplanet candidates.

It combines the rigor of astrophysics with the power of machine learning in one intuitive web interface.



From Upload to Insight — in Minutes

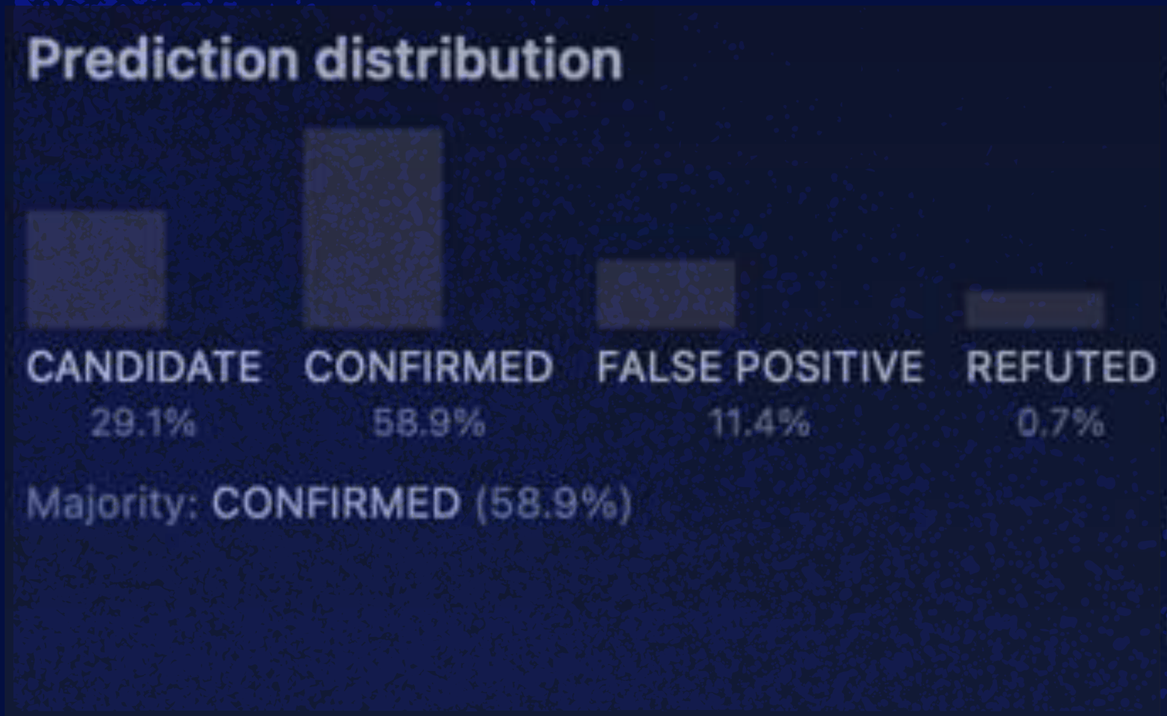
1. Upload your dataset (CSV or Excel)
2. Select the NASA mission (TESS, K2, or KOI)
3. Automatic preprocessing and normalization
4. Mission-specific AI classification
5. Results dashboard with metrics & visual insights
6. The model also uses the labelled datasets to improve its accuracy further

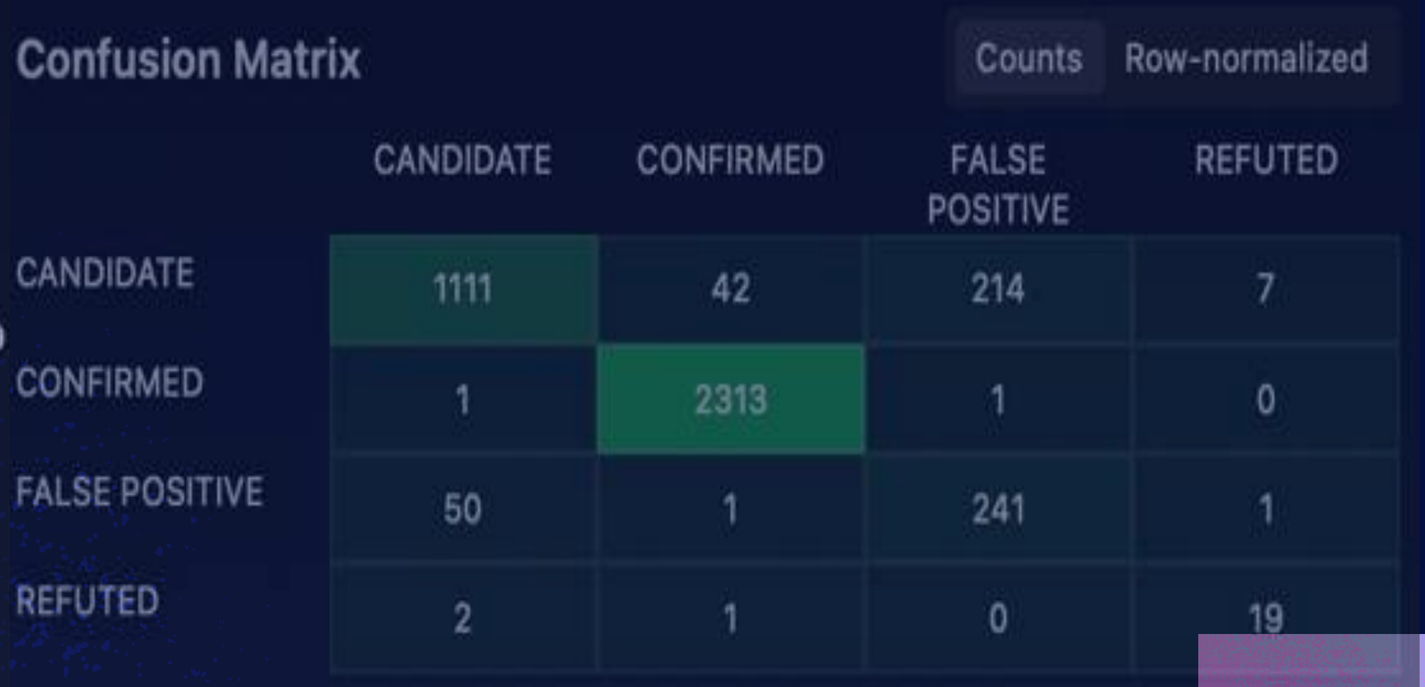
Mission-Aware Machine Learning

- Separate optimized ML models for Kepler, K2, and TESS
- Compared SVM, Logistic Regression, Random Forest, LightGBM
- Dynamic model selection via k-fold validation
- Class imbalance handled with weighted loss and tuning
- Transparency through auto-generated model cards

Accuracy Meets Explainability

- ExoVision processes over thousands of samples with mission-specific precision.
- Each evaluation includes a confusion matrix, class distribution, and bias analysis, helping scientists focus on discovery — not data wrangling.





Meet the Stellar Detectives Team

We are a passionate group of space enthusiasts and developers working together on the NASA Space Apps Challenge to explore exoplanets and bring data to life through innovation and design.



Saad Khan
Full-Stack Developer
3rd Year SENG



Muhammad Samad Mahar
UI/UX & Data Visualization
3rd Year PHY & CS



Abdullah Shah
Full-Stack Developer
3rd Year CS



Taimoor Shahzad
ML Research & Development
MEng ECE

A team united by curiosity — engineers, data scientists, and explorers.
Together, we built ExoVision to bring NASA’s open data to everyone.